Day 13 Report: RAG - Chat with Your Documents

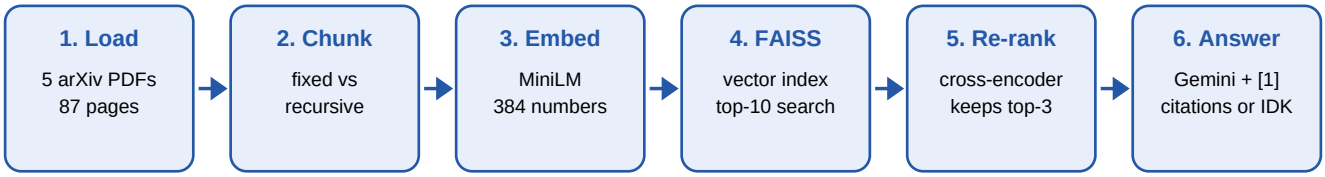
Retrieval-Augmented Generation over 5 AI research papers: chunking, FAISS, re-ranking, citations, evaluation

1. Goal

LLMs do not know private documents and can make up facts (hallucinate). **RAG** fixes this: first **find** the right passages in the documents, then let the LLM answer **only from them**, with **citations**, and say **"I don't know"** if the answer is not there. Think of it as an **open-book exam**.

2. How it works

RAG pipeline



Item	Value
Documents	Transformer, BERT, RAG, Sentence-BERT and LoRA papers (87 pages with text)
Chunk size	800 characters (fixed: 100 overlap; recursive: split by paragraph, line, sentence)
Embedding model	all-MiniLM-L6-v2 (384 dimensions), CPU
Vector store	FAISS IndexFlatIP (cosine similarity)
Re-ranker	cross-encoder/ms-marco-MiniLM-L-6-v2
LLM	Gemini (Colab built-in, no API key)
Test set	25 questions: 20 answerable from the papers, 5 not answerable

3. Chunking: fixed vs recursive

Strategy	Chunks	Avg length	How it splits
Fixed-size	468	717 chars	Every 800 characters, can cut a sentence in half
Recursive	434	685 chars	Paragraph > line > sentence > word, keeps sentences whole

4. Results: retrieval

Hit rate @3 = for how many of the 20 answerable questions at least one of the 3 chunks given to the LLM came from the right paper and contained the key fact.

Chunking	No re-rank	With re-rank
Fixed-size	55%	80%
Recursive	65%	80%

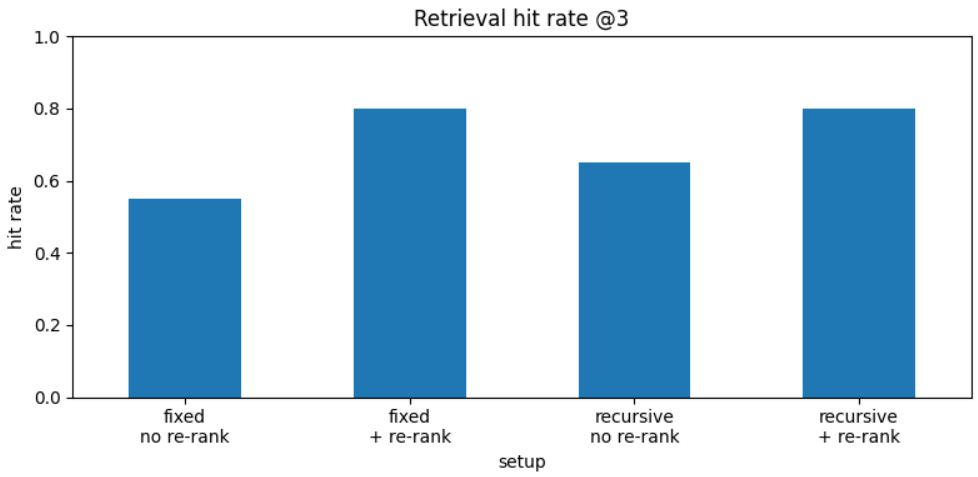


Figure 1. Retrieval hit rate @3 for the four setups

- **Re-ranking helped the most:** +25 points for fixed and +15 points for recursive chunks.
- **Recursive chunking was better without re-ranking** (65% vs 55%) because chunks keep whole sentences, so their meaning is clearer.
- With re-ranking both reached 80%, so the re-ranker made up for the weaker chunks. Recursive was used for the final answers.

5. Results: answers

Metric	Result
Retrieval hit rate @3 (recursive + re-rank)	80% (16 of 20)
Answerable questions answered	80% (16 of 20); 4 wrongly said "I don't know"
Unanswerable questions: said "I don't know"	100% (5 of 5), no hallucination
Faithfulness (LLM judge, 1-5)	5.0 average
Relevance (LLM judge, 1-5)	5.0 average

Question	Answer given	Why
% of tokens BERT masks	I don't know	Right chunk not retrieved
Largest model LoRA used	I don't know	Right chunk not retrieved
Default SBERT pooling	I don't know	Chunk found, but the fact was not clear in it
LoRA parameter reduction	I don't know	Chunk found, but model was too careful
SBERT training data	"STSb ... NLI datasets"	Partly right, missed SNLI/MultiNLI names
Transformer optimizer	Adam (correct)	Right answer, but from a less ideal chunk

6. Key findings

- **The "I don't know" rule works:** 5 of 5 unanswerable questions (for example the Duovate leave policy or the 2022 World Cup) were refused, so the assistant did not invent answers.
- **Retrieval is the weak point.** When the right chunk is not found, the LLM correctly says "I don't know", but the user gets no answer. Better retrieval = better answers.
- **Re-ranking is worth it:** a small model, fast on CPU, and it raised hit rate to 80%.
- **The LLM judge was too generous:** it gave 5/5 to every answer, even the 4 wrong "I don't know" answers. LLM judges must be checked by hand; my manual check found 15-16 of 20 answerable questions correct.
- Every answer had citations like [1] with paper and page, so it is easy to verify.

7. Limitations and next steps

- Only 25 test questions; a bigger test set would be more reliable.
- Some evidence keywords are weak (for example "mean" is a common word), which can make the hit rate look better than it is.
- Try: retrieving more chunks (top-5 to the LLM), hybrid search (keywords + vectors), smaller chunks for facts and numbers, and a stricter judge (or RAGAS).
- This RAG index will be reused as a tool for the Day 14 agent and the Day 16 MCP server.